

The correlation wall: a judgment

PRIME.RDAGGER.CORRELATION_WALL_JUDGMENT.01 — round 469

Stefan Hamann

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Abstract

This is an adjudication note. It builds no probe, measures nothing new, and proves no new theorem; it restates sealed identities, rules on five proposed languages plus three of its own, and issues a resource directive. The three verdicts are **U1** = INTERNALLY EXHAUSTED (no admissible new language; the reason is not fatigue but that the residual object is, after the named bridges, equivalent to the full Weil criterion, and that every language surviving the falsification test of round 467 demands input of Riemann-hypothesis strength), **U2** = EXTEND AND REFRAME (the standalone package is the right external object, but must carry the r467/r468 wall anatomy, a warning label about its own strength, and a separate fixed-window prize), and **U3** = REDIRECT TO FIXED L (the core is closed for rounds; the resources move to an unconditional classical theorem at *fixed* support length, which sits behind three already-named and individually standard tasks). No RH claim. No anti-RH claim.

Grades. [E] exact finite identity or elementary proved lemma; [C] sealed finite census; [O] open. Nothing below is promoted, and no census is extrapolated. Round references **rN** point to the sealed campaign record.

1 The object, and the wall in one page

Fix a selected window W_k of the named sequence $a_k = 2^k$, $r_k = \lfloor \sqrt{k} \rfloor$, $m_k = k2^{r_k} - 1$ (r461, r463). Let μ_k, ν_k be the positive and negative parts of the folded signed measure built from the von Mangoldt comb and the archimedean kernel, let $p_{k,0}, \dots, p_{k,N_k-1}$ be the μ_k -orthonormal polynomials, let σ_k be the fixed smooth comparison comb, and let B_k be the standing budget with reserve 5/7. With $C_k = \mathcal{B}_k^T \mathcal{B}_k$, $(\mathcal{B}_k)_{yi} = \sqrt{\nu_k(\{y\})} p_{k,i}(y)$ and $u_k = b_k / \sqrt{B_k}$, $b_{k,i} = \int p_{k,i} d\sigma_k$, round 460 proved

$$q_k^\dagger = u_k^T (I - C_k)^{-1} u_k = \sum_j \frac{|\langle u_k, e_{k,j} \rangle|^2}{1 - \lambda_{k,j}} \quad [\text{E}], \quad (1)$$

and $R^\dagger(W_k) \succeq \frac{1}{2}I \iff q_k^\dagger \leq 1$. The target is $q_k^\dagger \leq 1$ for infinitely many k .

The measured wall. $\lambda_{\max}(C_k) \uparrow 1$ fast ($1 - \lambda_{\max} = 8.7 \cdot 10^{-5}$ at $k = 12$), while the squared overlap of u_k with the top mode is $1.5 \cdot 10^{-6}$ to $9.5 \cdot 10^{-3}$; the race itself stays in $[0.606566, 0.757783]$ for $k = 5, \dots, 12$ [C] (r459, r460), and interval certificates now cover $k = 5, \dots, 16$ [C] (r465, extended r468). Every bound that treats the data and the modes separately fails by two to five orders of magnitude: the raw resolvent norm grows $4.825 \rightarrow 4991.649$, the optimal two-band bound exceeds 1 from $k = 6$, Gershgorin is negative on every row (r460), and the exact-coordinate Abel chain overshoots by 91.703 to 119482.632 with the prescribed 10^3 stop gate firing on $k = 7, \dots, 12$ (r467).

The protective cancellation is real. Three independent controls (r467, r468): the top modes are low-frequency (share 0.926–0.989) and the smooth ARCH border aligns with its top mode at 0.995813–0.998970; a deterministic source-position scramble *raises* the dangerous normalized top overlap by a factor 28.5 to $5.3 \cdot 10^5$; and at $k = 10 \rightarrow 11$ the isolated new prime octave moves $q^\dagger$ by +0.2199 at fixed geometry while the total step is -0.01180 . The cancellation therefore lives in the joint dependence of u_k and of the eigenbasis of C_k on the same arithmetic data.

Why every pairing bound must fail (diagnosis, not a bound). Since $\int p_{k,i} p_{k,j} d\mu_k = \delta_{ij}$, the matrix C_k is the Gram matrix of $\{p_{k,i}\}$ in $L^2(\nu_k)$ and $I - C_k$ is the Gram matrix of the *signed* folded measure $\omega_k = \mu_k - \nu_k$ on $\mathcal{P}_{<N_k}$. Restating (A5) of the dossier and (1) in the dual variational coordinate,

$$q_k^\dagger = \frac{1}{B_k} \sup_{0 \neq p \in \mathcal{P}_{<N_k}} \frac{\left(\int p d\sigma_k \right)^2}{\int p^2 d\omega_k}, \quad 1 - q_k^\dagger = \frac{\det \mathcal{A}_k}{\det(I - C_k)} \quad [\text{E}], \quad (2)$$

with $\mathcal{A}_k = \begin{pmatrix} I - C_k & u_k \\ u_k^T & 1 \end{pmatrix}$. Both forms are one object. The supremum in (2) is attained at the ω_k -Riesz representer of the smooth functional $p \mapsto \int p d\sigma_k$ — a single vector in which the arithmetic and the smooth comparison are already fused. Every route in the campaign’s kill list replaces that one supremum by a *product of two* suprema (a data majorant times a mode majorant). The loss is exactly the alignment that the controls measure, which is why the observed slack (10^2 – 10^5) is the same order as the scramble factor (28.5 – $5.3 \cdot 10^5$). This paragraph is a diagnosis of the failure class; it is not an estimate and must not be counted as progress.

2 U1 — the language question

2.1 A falsification test for candidate languages

The r467 controls turn into a methodological gate.

Criterion 1 (Scramble sensitivity; necessary, methodological). *Let $\mathcal{M}$ be a method producing an upper bound $\mathcal{B}_k \geq q_k^\dagger$ from window data. If the hypotheses of $\mathcal{M}$ are invariant under the r467 source-position scramble (which preserves channel weights, border atoms, and dimensions, and changes only the arithmetic positions), then $\mathcal{B}_k$ is also a valid bound in the scrambled world, whose dangerous edge alignment is larger by 28.5 to $5.3 \cdot 10^5$. Such an $\mathcal{M}$ must therefore absorb that factor, and cannot deliver $\mathcal{B}_k \leq 1$ along the sequence while the measured value sits near 0.7.*

This is a necessary condition of methodological type, not a theorem: the scrambled $q^\dagger$ was not measured, only the alignment it feeds. Its force is nevertheless decisive as a filter, because *all* thirty-one closed routes of the dossier (DCCXLVI–DCCCXXX) plus the r467 Abel chain and the r468 octave perturbation are scramble-invariant. That single sentence explains a decade of the kill table in one stroke, and it is hereby made a build gate (Section 4).

2.2 What the target actually is

Observation 1 (The residual is not a lemma). *Round 461 proved, from nestedness of the admissible support spaces, that $\lambda_*(L) = \inf\{Q_W(h)/\|h\|_2^2 : \text{supp } h \subset [-L, L]\}$ is nonincreasing, so nonnegativity along any unbounded sequence L_k forces nonnegativity at every finite L . Conditional on the three named external bridges (density/continuity, explicit-formula identification, zeta interface) and the two internal identifications, the requested cofinal statement is therefore equivalent to the full Weil criterion, hence to RH. Unconditionally it is at least as strong as cofinal compact-window positivity of the program’s own three-channel functional.*

Observation 1 is not new — it is r461’s own conclusion and it is printed in the standalone package. What is new here is the *judicial* use of it. Question U1 asks for a language that proves the residual. By Observation 1 that request is, up to bridges the program itself classifies as “known mathematics”, a request for a proof of RH. A judge may not answer such a request by naming a language; a judge may only answer whether any candidate survives Criterion 1 at bounded cost. None does.

2.3 The candidates, individually adjudicated

Effort/success is a judge’s estimate on the campaign’s own scale: LOW = one to three rounds, HIGH = a research programme, CONTAINED = the candidate contains an already retained open problem of the dossier.

Candidate	Why it would keep the correlation	Where it dies	Scr. sens.	Verdict / cost
(a) self-consistency, fixed point (Brouwer, contraction on the coupled system)	the coupling lives inside the map, so no factorization is performed	there is no small parameter. The only exact such map is diagonalization itself, so a fixed point is what already exists: existence, not localization. A contraction needs $1 - \lambda_{\max}(C_k)$ bounded below; the census gives $8.7 \cdot 10^{-5}$ at $k = 12$ and falling (r460), so the natural map expands at the edge	no	DISMISSED; dressed restatement, r303 alias risk; LOW cost, zero yield
(b) transfer operator, thermodynamic formalism on the fold dynamics	Gibbs state and spectral gap come from one operator; decay of correlations is intrinsic	there is no hyperbolic generator: the fold is a finite cosine identification, not an iteration, and Λ is not a Hölder Gibbs weight for any known potential. Decisively, the limit object was already refuted: r468 measured intra-block $\Delta q^\dagger$ oscillating in $[-0.025, +0.034]$ with an active-octave weight that <i>grows</i> , i.e. OCTAVE_NOT_CAUCHY — a thermodynamic limit needs exactly the summable rate that does not exist. Mayer/Lewis–Zagier transfer operators attach to Selberg-type zetas, not to this window	no	DISMISSED; limit object refuted by r468; HIGH cost, zero yield
(c) microlocal, semiclassical (comb as oscillating symbol, edge modes as Lagrangian states, non-stationary phase)	non-stationary phase <i>is</i> a correlation statement; r467’s low-frequency modes versus rough u is literally a wavefront-set separation	to run it one needs the symbol of C_k and the asymptotics of the μ_k -orthonormal polynomials. r467 records that C_k is a dense Gram matrix, not a Jacobi or pseudodifferential operator; supplying the missing asymptotics <i>is</i> the dossier’s retained Riemann–Hilbert problem. Expressed back in the source, the required gain is power-saving cancellation for $\sum_n \Lambda(n)n^{-1/2+it}$, i.e. a bound on ζ'/ζ on the critical line: RH-strength input, not a lemma	yes	DISMISSED as a proof route, RETAINED AS THE CORRECT DIAGNOSTIC VOCABULARY; CONTAINED in RHP

Candidate	Why it would keep the correlation	Where it dies	Scr. sens.	Verdict / cost
(d) determinantal, Fredholm: $q^\dagger$ as one determinant, not a pairing	(2) is exact and keeps the fusion inside a single ratio of determinants	as an identity it is a reduction, and this note supplies it for free. To <i>estimate</i> the ratio one needs Heine/equilibrium-measure asymptotics for the signed folded measure — the retained RHP problem again — or moment-Gram majorants, which is the closed r404 route (residual 209). The discrete IKS attempt already failed to produce a prefactor turning the Casoratian into $\det S_N$ (dossier, critical-block section)	partly	NOT NAMED as a language; recorded in Section 1 as the diagnostic coordinate. A round that only measures it is forbidden (Section 4)
(e) probabilistic decoupling: Sarnak–Möbius disjointness, pseudorandom measures	it does not need to keep the correlation; it asserts the decorrelation that the controls measure	disjointness statements require two <i>independently given</i> sequences. Here the observable is an eigenvector of a matrix built from the same data, so the self-reference is structural, not technical. Quantitatively, available savings are logarithmic while the gap to beat is $8.7 \cdot 10^{-5}$ and shrinking geometrically; and at the strength needed these statements are themselves RH-adjacent	no	DISMISSED; HIGH cost, zero yield
(f) <i>judge’s own</i> : free probability, operator-valued subordination, any genericity or randomness hypothesis on $\mathcal{B}_k$	subordination handles u against $I - C$ in one functional equation	falsified, not merely unproved: the “free” or generic world <i>is</i> the scrambled world, and there the dangerous alignment is $28.5 - 5.3 \cdot 10^5$ times larger (r467). A language whose hypothesis is independence predicts the wrong direction of the measured effect	no	DISMISSED by Criterion 1
(g) <i>judge’s own</i> : Markov/Christoffel comparison $\omega_k \succeq c\rho$ against an explicit positive measure (Loewner minorant with margin)	it would replace the fragile signed energy by a robust positive one	it separates by construction, and the uniform- c version is empirically false: $\lambda_{\min}(I - C_k) = 1 - \lambda_{\max}(C_k)$ collapses. Any σ_k -adapted, N -dependent version is the correlation again. The language is also half-used and half-closed already (r425 kernel Loewner; r407 ν -rank-one sandwich)	no	DISMISSED; alias

Candidate	Why it would keep the correlation	Where it dies	Scr. sens.	Verdict / cost
(h) <i>judge's own</i> : any increment, additivity, or quantifier renormalization in k (block limits, density of good k , subsequence extraction)	—	closed twice: r460 (fixed- r blocks have exactly $2r+1$ members) and r468 (REDUCTION_ALIAS : a block-limit quantifier is the same ∞ in new clothes)	no	DISMISSED ; r303 alias by construction

2.4 Verdict U1

U1 = INTERNALLY_EXHAUSTED. No admissible new language is named. The reason is structural, not a lack of imagination: exactly one candidate (c) survives the scramble filter of Criterion 1, and executing it requires the asymptotics of the μ_k -orthonormal polynomials (the dossier's retained Riemann–Hilbert problem) together with power-saving cancellation for $\sum_n \Lambda(n)n^{-1/2+it}$; by Observation 1 the target it would serve is, modulo bridges the program calls known mathematics, RH itself. The internal core is therefore closed *for rounds*. It is not abandoned: it stays as a documented external problem (Section 3), and (2) plus Criterion 1 are what this round adds to it. No RH claim, no anti-RH claim, and no claim that the residual is unprovable.

Remark 1 (What would reopen the core). Three named events, and only these: a proved edge-symbol or equilibrium-measure asymptotic for μ_k from outside the programme; a scramble-sensitive bound on $\max_r |\sum_{i \leq r} u_{k,i}|$ that is derived from the source rather than measured after the basis change (the verbatim r467 gap); or an external theorem improving $\lambda_*(L) \geq 0$ past the certified window range, which would make the fixed- L lane (Section 4) collide with the cofinal one.

3 U2 — the externalization question

Ruling. **EXTEND_AND_REFRAME.** The standalone package `cofinal_weil_positivity.tex` is the right external object: it is self-contained, repo-free, graded, and it already carries the faithful fold, the three equivalent coordinates, the certified census, and a failed-approach section. In its present form it is nevertheless mis-framed in one respect and under-powered in another. It *asks for a lemma* whose own r461 nestedness argument makes it RH-equivalent, and it omits the only genuinely new content the programme produced this week: a measured, reproducible mechanism plus a filter that explains the entire kill list.

What to add (five items, in this order).

1. **A strength warning, on page one.** State Observation 1 explicitly: after the three bridges the requested statement is equivalent to the full Weil criterion. Offer it as a *precisely located finite-matrix form of RH*, never as a lemma below it. This protects the document from both readings that would discard it — naive, or an RH claim in disguise.
2. **The wall anatomy** (r467, r468) as one section: the three controls with their numbers (low-frequency share 0.926–0.989; ARCH alignment 0.995813–0.998970; scramble factor $28.5\text{--}5.3 \cdot 10^5$; isolated octave +0.2199 against total -0.01180), the extremal identity (2), and the diagnosis of Section 1.

3. **Criterion 1 as the attacker’s filter**, with the honest label that it is methodological. This is the highest-value paragraph for an external specialist: it says which methods are dead before any work is done.
4. **The compressed kill list**: the thirty-one closed routes plus r467 and r468, one line each with the killing number. Attacker gold, and it is what makes the document a contribution rather than a request.
5. **The fixed-window prize, as a clearly separated section**: the interval-certified base $k = 5, \dots, 16$ with its widths, and the *conditional* classical reading $\lambda_*(L_k) \geq 0$ up to $L_{16} = k \log 2/4 = 2.77$, printed together with the exact list of what must be closed to make it unconditional (Section 4, F1). Presented as a second, strictly weaker, finite target — because it is the part an outsider can actually help with.

Where to send it.

- *arXiv*, math.NT primary with math.CA cross-list, as a problem-and-anatomy note (not as a result). The kill list and the filter are the citable content.
- *Direct correspondence*, three groups: the author of the compact-window certificate preprint (same technology, complementary coordinate — they hold the classical side, this programme holds the deeper certified windows); workers on Weil positivity and the archimedean place; and specialists in discrete orthogonal-polynomial duality and Riemann–Hilbert asymptotics, for candidate (c)/(d), which is precisely the retained RHP problem.
- *A specialist mathematics forum*, with the question **stripped of all RH framing**: a Markov-type inequality between an explicit signed folded measure and an explicit smooth comb on a polynomial window, in the form (2). Judge’s estimate: this is the highest-yield channel, because it removes the single feature that makes competent readers stop.
- *The Lean community channel*, for the two named formalization tasks only (Section 4, F1(i)–(ii)).

The repository package stays the reproducible appendix; no external document may depend on it.

What must not be claimed. No RH claim and no anti-RH claim. The mechanism is not evidence for RH and must never be phrased as such. The finite census and the interval certificates are not progress toward RH; they are finite statements about finite matrices. The residual must not be called a lemma. The compact-window preprint must not be cited as a theorem. The classical improvement of $\lambda_*(L)$ must not be claimed before F1(i)–(iii) are closed — until then it is conditional and must be printed with its conditions in the same sentence. Nothing in the anatomy may be described as explaining “why RH is true”.

4 U3 — the resource directive

4.1 Core: empty

Because U1 is INTERNALLY_EXHAUSTED, no round may target $q_k^\dagger \leq 1$ for infinitely many k , in any coordinate. The core survives as an external problem only. There is no core contract.

4.2 Foundation, in priority order

F1 (first). The unconditional classical theorem at fixed L . This is the directive’s centre. The programme has spent its recent rounds on the cofinal statement, which Observation 1 shows to be RH, while a genuinely new *unconditional* classical result sits behind three tasks that are individually finite or standard:

- (i) close `SelectedReadQuadraticRepresentation` — the single named remainder of r464, a precise finite structure problem about the channel map under unbounded `GridElement.steps`;
- (ii) close `arch_gauss_mellin_digamma_identity` — classical digamma/Mellin analysis, r464 already collapsed the archimedean gap to this one identity; formalization work, preceded by a Mathlib survey;
- (iii) prove the density and continuity step *at fixed support length only*: dyadic step autocorrelations are dense in $\{h \in L^2 : \text{supp } h \subset [-L, L]\}$ and Q_W is continuous there. At fixed L this is standard analysis; it is the cofinal version, and only that, which is hard.

Payoff: for a *single* k the r465/r468 interval certificate becomes an unconditional statement $\lambda_*(L_k) \geq 0$. At the currently certified top, $k = 16$, that is $L_{16} = 2.77$ against the 0.8 of the current external certificate — an improvement of the certified compact-window range by roughly a factor 3.5, with no cofinal quantifier anywhere. Why first: it is the only place in the programme where existing rigorous certificates convert into a classical, citable, non-RH result, and it is entirely independent of the wall.

First-round contract (Foundation, not Core). One round, one worker at implementation tier, scope limited to F1(i): produce either a proof of `SelectedReadQuadraticRepresentation` or a sealed statement of the exact finite obstruction, with no new `sorry` and no change to any other file; deliverable is the Lean change plus one `rh/problem` note and an INVENTORY row; exit gate is the RH suite green and the sorry census honestly reported. Explicitly out of scope: any statement about infinitely many k .

F2 (parallel, cheap). Certificates in the classical coordinate. Reproduce the external $L = 0.8$ certificate independently, then certify $\lambda_*(L) \geq 0$ directly at $L_5 = 0.866434$ and upward using the r465 interval machinery *in the classical coordinate*, i.e. without the fold dictionary. This makes F1’s payoff dictionary-independent, and it is the one lane whose value does not depend on any open bridge at all.

F3. Spectral side, fixed- L form only. The Guinand–Weil identification needed to complete F1(iii). Formalize it at fixed L . Do not formalize the cofinal quantifier; by Observation 1 that is RH and does not belong in a formalization backlog.

F4. Lean hygiene, continuous. The `sorry` census stays honest at eight and decreases only by real closures. No new `sorry` without a round reference. Every closure gets an INVENTORY row and a `run_rh.py` gate in the same change.

Deliberately unfunded. The Riemann–Hilbert/steepest-descent route and the retained minimal-inertia problem stay documented and unfunded. They are the correct external asks (Section 3), not rounds.

4.3 External

One round to extend the package per Section 3, one round to dispatch it through the four channels. No probe in either.

4.4 Anti-list (binding; r303 discipline)

1. No further majorant, operator-norm, Gershgorin, triangle, Abel/summation-by-parts, or two-band variant. The class is exhausted, and its failure is now explained by Criterion 1 rather than merely recorded.
2. No restatement of the ∞ -quantifier: block limits, octave Cauchy rates, density of good k , plateau arguments, subsequence extraction. Alias by construction (r460, r468, r303).
3. **Mandatory scramble gate.** No probe may be built until its proposed bound is declared scramble-sensitive, with the reason written down. A scramble-invariant proposal is refused before any code is written.
4. No round on candidates (a), (b), (e), (f), (g), (h) of Section 2 without first exhibiting the missing hypothesis (small parameter, Gibbs potential, independence, uniform minorant) and showing that it survives item 3.
5. No round that requires the RH-asymptotics of the μ_k -orthonormal polynomials. That is the retained RHP problem: an external ask, not a round.
6. No further anatomy of the same wall without a named bound target in the round contract. Anatomy is cheap and it is now sufficient; bounds are what is missing.
7. No RH claim, no anti-RH claim, no “evidence” language, and no promotion of any census or interval certificate to a statement about ζ .

Claim boundary. This note proves nothing new. Its identities (1) and (2) are restatements of sealed results (r460 and the dossier’s augmented dictionary, item A5); its numbers are quotations from sealed rounds; Criterion 1 is explicitly methodological and not a theorem; Observation 1 is r461’s own conclusion. Nothing here is evidence for or against the Riemann hypothesis. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

References

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